PROJECT DOCUMENTATION Online Coding Platform

1️⃣ Project Title Online Coding Platform for Coding Practice and Assessment

2️⃣ Abstract

The Online Coding Platform is a web-based application that allows users to practice coding problems, participate in coding contests, and improve their programming skills. The platform supports multiple programming languages and provides real-time code execution and evaluation. It also includes an admin panel to manage problems, users, and submissions.
The goal of this project is to provide an efficient and user-friendly system similar to platforms like LeetCode and HackerRank.

3️⃣ Problem Statement Many students struggle to find a structured platform where they can:

Practice coding problems
Get instant feedback
Participate in contests
Track performance
Existing platforms are either paid or too complex. This project aims to create a simple, accessible, and efficient coding platform for students.

4️⃣ Objectives

✅ Provide coding problems with different difficulty levels
✅ Support multiple programming languages
✅ Real-time code compilation & execution
✅ Store user submissions
✅ Admin dashboard to manage problems
✅ User authentication & authorization

5️⃣ Technologies Used Frontend:

HTML
CSS
JavaScript
React.js (if used)
Backend: Node.js
Express.js
Database: MongoDB
Other Tools: Judge0 API / Code execution API
JWT Authentication

PROJECT DOCUMENTATION Online Coding Platform

Git & GitHub

6️⃣ System Architecture User (Frontend) ↓ React UI ↓ Backend (Node.js + Express) ↓ Database (MongoDB) ↓ Code Execution API

7️⃣ Modules of the Project

- 1. User Module Register

Login
Logout
View problems
Submit code
View submission history

- 2. Admin Module Add new problems

Edit/Delete problems
View users
Manage submissions

- 3. Code Execution Module Accepts code

Sends to compiler API
Returns output
Shows test case result

8️⃣ Database Design Users Collection

UserId
name
email
password (hashed)
role
Problems Collection problemId
title
description
difficulty
testCases
Submissions Collection submissionId
userId
problemId
code
language
status
timestamp

PROJECT DOCUMENTATION Online Coding Platform

9️⃣ Key Features

🔐 Secure login using JWT
⚡ Real-time code execution
📊 Submission tracking
🏆 Difficulty-based problem filtering
👨‍💻 Multi-language support
📱 Responsive UI

🔟 Challenges Faced Secure code execution

Handling asynchronous API responses
Managing user authentication
Preventing infinite loop execution

1️⃣1️⃣ Future Enhancements

🏆 Leaderboard system
🧠 AI-based code suggestions
📊 Performance analytics dashboard
💬 Discussion forum
📱 Mobile app version

1️⃣2️⃣

Conclusion

The Online Coding Platform successfully provides a structured environment for students to practice coding and improve problem-solving skills. It demonstrates strong understanding of full-stack development, API integration, authentication, and database management.